

A Hybrid A* Path Planning Algorithm Based on Multi-objective Constraints

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Abstract—To provide a safe, smooth and efficient global planning path for nonholonomic mobile robots. Aiming at the problems of traditional hybrid A* algorithm in path planning, such as approaching obstacles, unnecessary reversing and redundant turning, a multi-objective constraint method based on hybrid A* algorithm is proposed. To speed up the path planning, the heuristic function is dynamically weighted and the overall path cost function is designed. The path planning experiments are carried out in ROS (Robot Operation System) simulation environment and actual environment respectively, and the results show that. The hybrid A* algorithm with multi-objective constraints increases the minimum distance between the robot and obstacles by more than 50%, reduces the unnecessary times of reversing and turning, and reduces the total running time by 14.2% on average, thus improving the navigation efficiency of the mobile robot.

Keywords—nonholonomic constrained robot, multi-objective constraints, hybrid A*, path planning

I. INTRODUCTION

Path planning as a basic function of autonomous motion of mobile robots, its task is to find a driving path from the current point to the target point so that the mobile robot can reach the target location safely and quickly in this task [1]. Its algorithms are mainly random sampling-based search algorithms and graph-based search methods. Sampling-based algorithms have the characteristics of fast search speed but poor handling of complex environments, and the representative algorithms are the probabilistic sign method PRM [2] and the fast search random tree method RRT [3]. A* algorithm [5] is a global path planning algorithm that combines the heuristic function of Dijkstra's algorithm to quickly plan a global route based on the shortest path. However, as a traditional graph search approach is not applicable to non-integrity constrained vehicles such as Ackermann steering. To address this problem, Dolgvo [6] et al. proposed a hybrid A* algorithm based on A* and satisfying the kinematic properties of the vehicle, but generating large curvature changes and not smooth. Chu et al

[7] used PH triple curves for post-processing of the path based on the hybrid A* algorithm search path, but ignored the distance problem with obstacles; Kurzer [8] et al. proposed A* with Reeds-Shepp curves alternately for node expansion, which speeds up the search, but there are problems such as the robot being too close to obstacles and redundant steering in the planned paths. Tang [9] et al. proposed the artificial potential field method as an optimization method for hybrid A*, but the traditional artificial potential field method constructs a high potential field in narrow sections, which is not conducive to robot passage in the case of dense obstacles.

To address the above existing problems, we propose a hybrid A* algorithm based on multi-objective constraints. The algorithm proposes the cost function of multi-objective constraint about the distance between reversing, steering and obstacle. It also combines the idea of weighted A* to speed up the search of paths. Finally, further smoothing of the generated node paths is performed. The improved hybrid A* algorithm improves the safety, smoothing, and reduces the planning time of path planning, thus improving the overall navigation efficiency of the robot.

II. MOTION ANALYSIS AND ALGORITHM IMPROVEMENT

A. Movement Analysis

The motion analysis is first performed for a non-complete constrained robot with Ackermann structure. According to Ackermann's principle [10], such robots have corresponding geometric constraint relations in the path planning process, such as the inability to move sideways and the existence of a minimum turning radius.

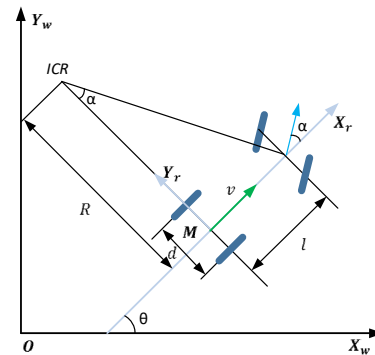


Fig. 1. Ackerman chassis motion model

Establish the world coordinate system as X_wOY_w , and the local coordinate system of the robot as X_rMY_r . The global coordinate system is mapped into the local coordinate system using the orthogonal matrix. θ is the deviation angle between the global and local coordinate systems, ICR is the instantaneous center of rotation of the chassis. The turning radius α and the steering angle have the geometric relationship as

$$R = l / \tan \alpha \quad (1)$$

where R denotes the instantaneous turning radius, l is the front and rear axle distance. The relationship between angular velocity and linear velocity as

$$w = v / R \quad (2)$$

Where v is the rear-axis center linear velocity, w is the rear-axis center angular velocity. In time, the two adjacent positional movements can be approximated as an arc motion with constant curvature. The kinematic equations of the Ackermann chassis are established as

$$\dot{s}(t) = \begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} v(t) \cos[\theta(t)] \\ v(t) \sin[\theta(t)] \\ v(t) / [l \cdot \tan \alpha(t)] \end{bmatrix} \quad (3)$$

Where $\dot{s}(t)$ denotes the derivative of the attitude state, $v(t)$ is the linear velocity at a certain moment, $\theta(t)$ is the deviation angle between the world coordinate system and the local coordinate system at a certain moment, $\alpha(t)$ is the steering angle at a certain moment. The relationship between the steering angle and the angular velocity of the chassis linear velocity as

$$\alpha(t) = \arctan[w(t) \cdot l / v(t)] \quad (4)$$

The input control quantity for the Ackermann robot motion is $u(t) = [v(t), \alpha(t)]$, and there is a corresponding range of steering angles due to mechanical constraints.

B. Hybrid A* Algorithm

The A* algorithm extends nodes in discrete space from the centroid of the current grille to the surrounding eight grille centroids, as shown in Figure 2(a). The path is a concatenation of two-to-two centroids that retrace the expanded grids and thus generate the path, the path expansion approach of A* search does not consider the vehicle kinematic constraints and does not satisfy the non-complete constrained robot of Ackermann steering. The hybrid A* algorithm [11] incorporates the kinematic constraints of the vehicle in the traditional A* algorithm, and the search is shown in Fig. 2(b). Each successive state of the algorithm is given by (x, y, θ) , which is the position and direction of the vehicle, respectively. During the search, the search space is discretized and the forward motion is expanded by one of three actions: maximum left steering action, maximum right steering action, and no steering, and the same for reversing.

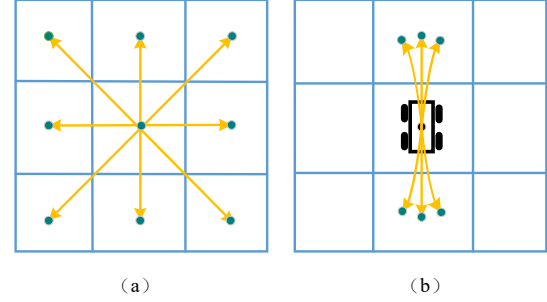


Fig. 2. A* and hybrid A* node search (a) A* Search (b) Hybrid A* Search

Two heuristic functions are used in parallel in hybrid A* in order to satisfy the constraints of its own kinematics and environmental obstacles [9].

The first heuristic function considers the non-integrity constraint of the vehicle, ignoring the obstacles, and its value is the shortest path that satisfies the non-integrity constraint in the current node $(x_{node}, y_{node}, \theta_{node})$ reaching the target point $(x_{goal}, y_{goal}, \theta_{goal})$, which is obtained by the Reeds-Shepp curve. This curve calculates the distance from the starting point to any point offline and obtains the heuristic value from any point to the target point by rotating and translating the real-time 3D vector.

The second heuristic function is obtained by considering the obstacle environment and ignoring the vehicle kinematics. The heuristic function is performed in two dimensions and the heuristic function values are obtained by the A* algorithm. The larger value of the two heuristic functions is chosen as the heuristic for the hybrid A*. The heuristic function as

$$h(s_i) = \max[h_1(s_i), h_2(s_i)] \quad (5)$$

Where $h_1(s_i)$ denotes the value of the heuristic function considering the vehicle non-integrity constraint, $h_2(s_i)$ denotes the value of the heuristic function considering the obstacle environment. The path expansion in the first heuristic function considers the kinematic constraints of the vehicle, but the path contains unnecessary steering actions, which reduces the robot's execution efficiency. The second heuristic function, which can guide the vehicle to avoid dead ends and U-shaped obstacles, but the obtained path tends to approach the obstacles in pursuit of the shortest path, poses a threat for the safety of the robot.

C. Improved Hybrid A* Algorithm

In this paper, we add multi-objective constraints on the paths to the hybrid A* algorithm to make the generated initial paths more smooth and secure, and construct a new overall cost function using dynamic weighting for the heuristic function to speed up the search efficiency of the paths. Conjugate gradient smoothing is performed on the generated initial paths, and the optimized planning route is output. The improved algorithm flow is shown in Figure 3.

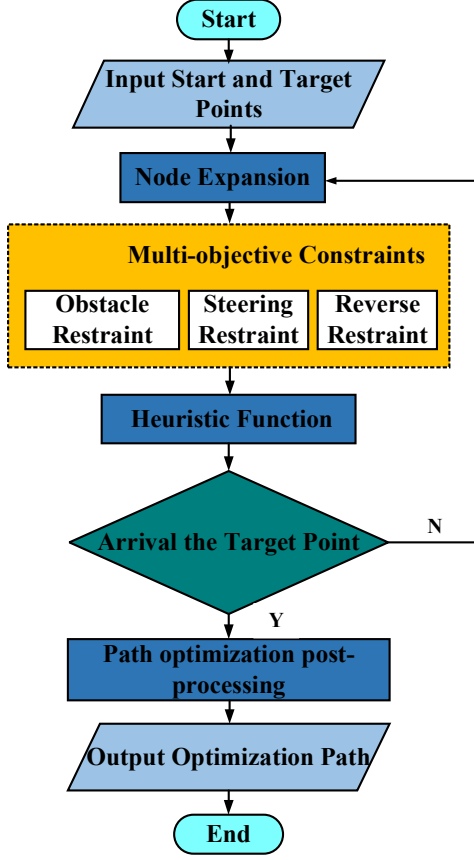


Fig. 3. Flow chart of improved algorithm

a. Multi-objective Constraints

For different obstacle scenarios, in order to make the improved path planning meet the requirements of smooth, safe and less reversal. Add the constraint cost of multi-objective, the extended cost function from node s_{i-1} to the next node s_i as

$$cost(s_{i-1}, s_i) = d(s_{i-1}, s_i) \cdot \left(1 + \frac{|\delta_{i-1}|}{2} \cdot w_r + \frac{|\delta_i - \delta_{i-1}|}{2} \cdot w_t\right) + risk_i \quad (6)$$

Where $d(s_{i-1}, s_i)$ denotes the path length of two adjacent nodes, δ_i is the vehicle driving state at the i^{th} node, $\delta = 1$ representing forward driving and $\delta = -1$ reverse driving, w_r is the cost coefficient of reverse driving, w_t is the cost coefficient of steering, $risk_i$ is the penalty term of distance from the obstacle at the i^{th} node, and the expression as

$$risk_i = \begin{cases} w_o \cdot (d_{min} - d_{ob}), & d_{min} \geq d_{ob} \\ 0, & d_{min} \leq d_{ob} \end{cases} \quad (7)$$

Where w_o is the cost coefficient of obstacle at less than the safe distance, d_{min} is the minimum safe distance allowed from the obstacle, d_{ob} is the current node distance from the obstacle. When there are reversing, turning and approaching obstacles in the scene, the multiplication term of the cost

coefficient is not zero and the overall cost function increases. When w_r infinitely large, it means that the current algorithm does not allow reversing.

b. Heuristic Function

In order to speed up the path search efficiency, the above improved cost function and the heuristic function of the traditional hybrid A* algorithm are used to construct the overall path function in a dynamic weighted manner. The function is designed as

$$f(s_i) = g(s_i) + \left(1 + \frac{d_1}{d_2}\right) \cdot h(s_i) \quad (8)$$

$$g(s_i) = \sum_{n=1}^i cost(s_{n-1}, s_n), \quad i = 1, 2, \dots, n \quad (9)$$

Where $f(s_i)$ denotes the overall generation value of the path, $g(s_i)$ denotes the cumulative generation value from the starting point to the current node, $h(s_i)$ is the maximum inspired value in equation (5), d_1 is the distance from the current node to the target node, d_2 denotes the distance from the starting node to the target node. At the starting point, the body position requirement is low, and the node expansion is performed at the expense of the optimal path, with a heuristic function $h(s_i)$ weight of 2, and the fastest search speed. When the robot approaches the end point, factors such as parking orientation will require higher body position of the robot, so the heuristic function $h(s_i)$ at the end point has a weight of 1, and the search speed is the slowest, which tends to select the optimal path.

c. Path Optimization Post-processing

In order to make the initial path of the planning smoother, the conjugate gradient method is used for path optimization in this paper [12]. The objective function established as

$$P = w_o \sum_{i=0}^N (|X_i - O_i| - d_{min})^2 + w_c \sum_{i=0}^{N-1} (k_i - k_{max})^2 + w_s \sum_{i=0}^{N-1} (\Delta X_{i+1} - \Delta X_i)^2, \quad i = 1, 2, \dots, N; \quad (10)$$

Where the first term represents the obstacle constraint term, the second term represents the curvature constraint term of each path point, and the third term represents the path smoothing term. w_o , w_c , w_s are the weights occupied by each function, and the size of the weights is obtained by experiment, the obstacle term is to limit the distance between the path point and the obstacle to avoid collision, X_i is the coordinates of the path node, O_i is the coordinates of the nearest obstacle, d_{min} is the minimum safe distance from the obstacle, the curvature term is to constrain the curvature of any path point, k_i is the curvature of any node, and k_{max} is the maximum curvature that can be achieved by the intelligent vehicle model to ensure the driveability of the optimized path. The smoothness term $\Delta X_i = X_i - X_{i-1}$ denotes the direction vector composed of two adjacent nodes, which is penalized by the direction vector changes of all nodes in the path, thus reducing steering and improving the smoothness of the path.

III. RESULTS AND DISCUSSION

The hybrid A* algorithm with node expansion approach and multi-objective constraints proposed in this paper is validated in different scenarios through simulation and real vehicle experiments. The planning time, number of turns, number of reversals, obstacle distances, and route execution time are compared. The relevant parameters of the experiments are shown in Table 1.

Table I. Test Related Parameters

Parameters	value
Vehicle Wheelbase l (mm)	275
Wheelbase d (mm)	200
Maximum front wheel steering angle α (°)	30

A. Path Planning Simulation Experiments

Simulation is performed in ros environment to build Ackermann steering trolley with parameters as above, and Rviz is used as a visualization tool. The path planning experiments are conducted in low-density scenario 1 and high-density scenario 2 to compare the algorithm before and after improvement, as shown in Figure 4 and 5. The green arrows reflect the position and orientation of the starting point of the cart, the green curve represents the algorithm planning path, and the red area represents the area swept by the body in the map.

Scenario 1 Simulation Experiment.

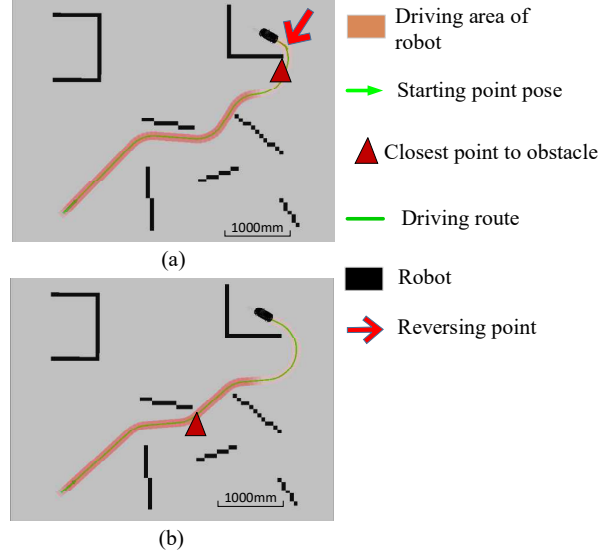


Fig. 4. scenario 1 (a) Hybrid A* (b) Improved hybrid A*

As shown in Figure 6, the improved hybrid A* algorithm is compared with hybrid A*. The improved algorithm reduces the planning time by 19.9%, the number of reversals by 1, the minimum distance from the obstacle in the path by 103.2%, and the total time (total time = planning time + execution time) by 13.8%. The specific parameters are shown in Table 2 below.

Table II. Scenario 1 Path Parameters

	Plann-ing time (ms)	Steer-ing times	Rever-sing times	Minimum distance to obstacles (mm)	Path execution time (s)
Hybird A*	367.54	4	1	31	29
Improved hybrid A*	294.53	4	0	63	25

Scenario 2 simulation experiment.

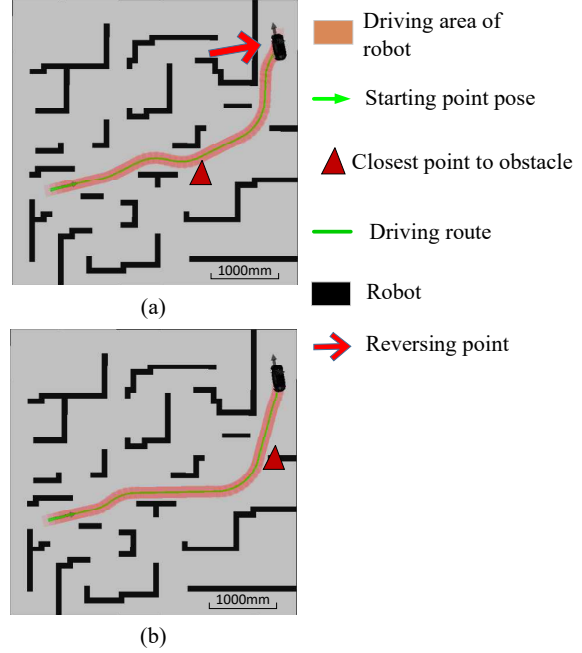


Fig.5. scenario 2 (a) Hybrid A* (b) Improved hybrid A*

As shown in Figure 7, the improved hybrid A* algorithm reduces the planning time by 30.0%, reduces the number of turns and reversals by 1 each, increases the minimum distance to the obstacle by 59.4%, and reduces the total time (total time = planning time + path execution time) by 17.5%. The specific parameters are shown in Table 3 below.

Table III. Scenario 2 Path Parameters

	Plann-ing time (ms)	Steer-ing times	Rever-sing times	Minimum distance to obstacles (mm)	Path execution time (s)
Hybird A*	302.01	5	1	32	23
Improved hybrid A*	211.28	4	0	51	19

In simulation experiments, it is shown that the improved hybrid A* can calculate relatively smooth paths more quickly in different scenarios, increase the minimum distance to obstacles by more than 50%, reduce unnecessary reversing and steering actions, and ensure the executability and execution efficiency of the planned routes.

B. Path Planning Real-world Experiments

RTAB-SLAM is used for the actual scene construction, and the path planning plug-in is registered in the move_base

navigation framework of the path planning module in the upper computer ROS. Hybrid A* algorithm and improved hybrid A* algorithm are used in global path planning, and TEB (Timed Elastic Band) algorithm is used for local path planning, respectively. In the real vehicle experiment, the robot can complete the autonomous path planning task based on the a priori environmental obstacle information, and the trajectory route is shown in Figure 6.

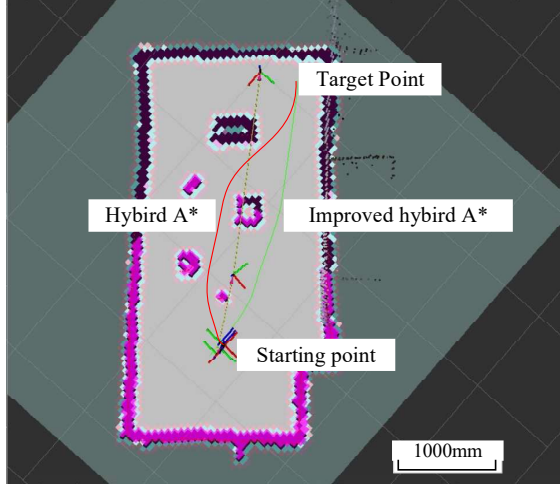


Fig.6. Real robot planning route

To verify the effect of steering, reversing and obstacle distance on the execution efficiency of the hybrid A* algorithm before and after the improvement, the actual execution process is analyzed and compared, as shown in Figure 7.



(a)



(b)

Fig. 7. Real robot test of route planning

(a) Hybird A* (b) Improved hybrid A*

The interval between adjacent positions of the robot in the figure is 3s, and each parameter is shown in Table 4 below.

Table IV. Real Robot Test Path Parameters

	Planning time (ms)	Steering times	Reversing times	Minimum distance to obstacles (mm)	Path execution time (s)
Hybird A*	366.56	4	0	90	27
Improved hybrid A*	290.04	2	0	122	24

In the real scenario, the improved hybrid A* algorithm selects the area with fewer obstacles to pass compared to the hybrid A*. And the planning time of the improved algorithm is reduced by 20.9%, the minimum distance from the obstacles in the path is increased by 57.5% and the total time (total time = planning time + execution time) is reduced by 11.2% for the traditional hybrid A* algorithm reversing once. It is further verified that the improved hybrid A* algorithm has faster and better global path planning routes under the constraint of multiple objectives.

IV. CONCLUSION

A multi-objective constraint-based hybrid A* algorithm is proposed for the problems of the hybrid A* algorithm in global path planning, and the following conclusions are drawn from the experimental study.

A control model with Ackermann incomplete motion constraints is established. Obstacle constraints, reversing constraints and steering constraints are introduced into the known cost of path node expansion to increase the minimum distance to obstacles, reduce the number of unnecessary reversing and steering, and improve the safety and smoothness of the planned path. The dynamic weighting of the heuristic function improves the path planning speed. The improved algorithm is experimentally verified in simulation and physical scenarios. The results show that the improved hybrid A* algorithm tends to be smoother, reduces the number of non-essential turns and reversals, and maintains a suitable distance from obstacles. The improved hybrid A* algorithm reduces the total running time by 14.2% in three scenarios and improves the efficiency of mobile robot navigation.

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